Title

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Introduction

Prostate carcinoma (PCa) is the second most prevalent form of cancer among men worldwide accounting for 29.4% of cancer cases and is the fifth leading cause of cancer deaths among men (Bray et al. 2024). In spite of this, the most recent Cochrane [This is a pretty old review from 2013. Consider citing some newer recommendations] review of prostate cancer screening using prostate-specific antigen (PSA) testing and digital rectal examination (DRE) showed that there was no difference in prostate-cancer specific mortality between screening and control groups. Furthermore, there was a significant increase in moderate-severity harms due to screening including anxiety and bleeding as well as the risk of false-positives with over-diagnosis and over-treatment (Ilic et al. 2013). Therefore, there is a need for both less-invasive and more-accurate screening methods to improve the effectiveness of early screening and to reduce the negative consequences for screening among men. Furthermore, there is a demand for identifying individuals at elevated risk of aggressive, metastatic, or clinically-significant prostate cancer for early screening of prostate cancer via PSA and MRI to avoid over-diagnosis and over-treatment of indolent disease.

One potential approach to reduce the false positive rate for early screening and to reduce the risk of harms due to DRE is to employ genetic screening to stratify individuals based on cancer risk. The most common approach to genetic screening for cancer is with rare pathogenic variants (RPVs). For example, women carrying a pathogenic mutation in the BRCA2 gene have a 60% chance of developing breast cancer in their lifetime [citation]. There are a number of RPVs associated with increased risk of PCa that are, in some cases, associated

with worse prognosis, including BRCA1, BRCA2, and HOXB13 G84E [cite Tyler's paper]. Carrying one of these variants can increase an individual's risk by factor of 1-8.5 [Tyler]. However, these variants are at relatively low frequency in the population (typically, 0.05-0.10%) and therefore are of limited use in stratifying most people, though certain variants such as HOXB13 are at a relatively high frequency in Scandinavian populations of 1.4% (Seibert).

An alternative approach to screening for RPVs is to use polygenic scores (PGS). In this case, many common SNPs with small effect sizes are combined to calculate a PGS for an individual. Individuals with high (2-3x) or very high (3.5-9x) risk based on PGS—comparable in magnitude to carrying a single RPV—make up as much as 20-25% of the population. Therefore, PGSs can potentially be more useful for identifying many more individuals at increased risk of cancer to target for further screening. Furthermore, screening for individuals using genotyping is much cheaper than alternatives such as whole-genome sequencing.

[One of the arguments for genotyping is that it is much cheaper than whole-genome sequencing, but if you are screening individuals for, for example, a BRCA2 mutation, you wouldn't WGS them, you would probably do a targeted sequencing panel, so I am not sure how strong of an argument that is.]

Typical PGS models use case-control designs with logistic regression to calculate a polygenic risk score (PRS), to estimate the risk of developing cancer at some point in their lifetime. More recently, investigators have begun using information about age of diagnosis or age of onset in a survival modeling framework to estimate the risk of developing disease at an early age. Genetic risk scores using this survival modeling framework are called polygenic hazard scores (PHS) and have been demonstrated to stratify individuals based on predicted age of onset for prostate cancer in multi-ethnic groups (Seibert or another PHS citation, Huynh-Le et al. 2021 [cite the PHS601 paper]). The most recent PHS for PCa has also been shown to be sensitive/specific enough to identify individuals at greater risk of metastatic/aggressive disease. Furthermore, PHS can be combined with PSA screening and family history to potentially improve the positive predictive value of these screening measures (Seibert et al. 2018). By stratifying individuals using PHS, we can identify individuals who would benefit most from early PSA screening. Stratifying individuals by PHS score could also improve future pharmaceutical studies by identifying individuals with the highest risk of developing prostate cancer.

Currently, few studies have tested published PHS models in populations that differ from the development set. One example of this was the PHS model of Alzheimer's disease from Desikan et al. (2017), which was tested in novel Nordic cohorts from Norway and Denmark (Motazedi 2023). While several PHS models for prostate cancer age-of-diagnosis have been published (Seibert, others), to our knowledge, none have been externally validated in populations that may have missing or imputed SNPs from the model. The earliest models in this area were criticized for being developed in European populations and for not being generalizable to non-European populations. However, more recent PCa PHS models have been developed in more diverse cohorts, such as the Million Veterans Program in the US, and have been shown to stratify individuals in multi-ethnic populations (Huynh-Le et al., 2021, Seibert, 2025 pre-print).

In this study, we evaluated the latest published PHS model for individual-level early diagnosis of PCa by performing external validation in two Nordic populations. In addition, we performed whole-genome sequencing on a subset of individuals to identify RPVs and compare the efficacy of screening using PHS and RPVs. Lastly, we validated whether the PHS601 model is sensitive enough to identify individuals at increased risk of metastatic disease.

Norway offers a unique opportunity for investigating these questions because of their well-developed electronic health records system, which allows investigators to link patient data from disparate health registries including the cancer registry and the death registry. This provides comprehensive information on

Methods

Subjects/Study Population

Two Norwegian cohorts were included in this study. The first we will refer to as the Norwegian Familial Cancer Study (NFCS). Subjects in this cohort were recruited in 1980-1995 from all of Norway via self-report based on over-representation of cancer in their families. Genomic DNA was extracted from peripheral blood samples using the DNeasy Blood & Tissue Kit (Qiagen, Germantown, MD, USA), following the manufacturer's protocol, and analyzed using the Illumina OmniExpress 24 v1.1 chip at deCODE genetics (Reykjavik, Iceland) [cite Mattingsdal]. The genotyped dataset consisted of 15,769 subjects with 713,014 SNPs. Genotype data were imputed using a reference panel from The Norwegian Mother, Father and Child Cohort (MoBa) with the MoBa pipeline [cite Corfield et al. I guess]. After imputation, the dataset consisted of 7,526,252 SNPs.

The second cohort was the Hordaland Health Studies (Helseundersøkelsene i Hordaland) (HUSK). [How was HUSK genotyping and imputation performed? Using the MoBa pipeline, I presume, but which reference panel was used?]

Prior to analysis, each dataset was filtered to remove female subjects, subjects under age 40, and subjects lacking genotype data.

Characterization of PCa Cases

Metastatic disease was defined as disease prevalence at the time of diagnosis that extends to lymph nodes inside or outside the same body region, organ metastasis outside the body region, and microscopic or macroscopic ingrowth into neighboring structures.

Clean summary statistics and PHS calculation

Summary statistics from previously published PCa PHS models were acquired from the PGS Catalog or by contacting the authors. Each summary statistics file was pre-processed with the cleansumstats pipeline to harmonize effect and reference alleles as well as chromosome

positions and RSIDs with revision 37 of the 1KG reference panel [reference?]. This pipeline removed palindromes, indels, and impossible allele pairs and removed SNPs from the major histocompatibility complex (MHC) (cite this). Both summary statistics for the models and genotype data for the subjects were processed through cleansumstats. After the pipeline, of the initial 601 SNPs in PHS601, 109 SNPs were removed as palindromes, indels, or impossible allele pairs leaving 492 SNPs in the model. In addition, the genotype files from the two cohorts were processed through the cleansumstats pipeline to insure that RSIDs and effect alleles matched between the models and the genotype data. Of the 492 SNPs remaining from PHS601, 41 SNPs were missing from the NFCS genotype data, which meant the PHSes for these subjects were calculated from the remaining 451 SNPs. Likewise, ## SNPs were missing from the HUSK genotype data so the PHSes for these SNPs were calculated from ## SNPs remaining from PHS601. Polygenic hazard scores were calculated for each model-cohort combination with matrix multiplication. The resulting PHSs were centered and scaled prior to modeling.

Sequencing

Information on number of subjects with wgs.

Variant calling

Variant calling was performed using the CPSR package [Nakken et al. 2021]. We identified individuals with 1 or 2 copies of pathogenic variants known to be associated with incrased risk of PCa including mutations within the genes ATM, BRCA1, BRCA2, HOXB13, MLH1, MLH2, and MSH6.

Statistics

Polygenic hazard scores were evaluated using Cox proportional hazards models using the `coxph` function from the `survival` package in `R` (cite). For the any-PCa model, events were defined as age of first PCa diagnosis and controls were censored at age of last observation. For the metastatic PCa model, events were defined as age at first metastatic disease diagnosis. Polygenic hazard scores were centered and scaled prior to model fit. After model fitting, subjects were separated by centile range of PHS into the top 20%, the bottom 20%, and the middle 60% for plotting of K-M curves. Performance of each model is reported as $HR_{80/20}$, which is the hazard ratio between the top 20% and the bottom 20% as well as $HR_{80/60}$, which is the hazard ratio between the top 20% and the middle 60%. For these metrics, confidence intervals were generated using 999 bootstrap iterations.

To test whether RPVs were at higher frequency in cases than in controls, we used the Fisher's exact test. Two genes for which no variants were observed in our cohort (MSH2 and MSH6) were removed prior to analysis. The remaining 5 genes were tested individually as well as aggregated as individuals carrying any of the variants. The p-values from these tests were adjusted using the Benjamini & Hochberg correction (1995). Genes at elevated

frequency in the cases were then used in a Cox survival analysis to estimate the risk of PCa associated with carrying one of these pathogenic variants. To test whether individuals with high PHS were more likely to carry a rare variant, we used logistic regression. All Cox models were evaluated for the proportional hazards assumption using the `cox.zph` test in `R` and by inspecting plots of the Schoenfeld residuals versus time via `plot.cox.zph` (Cite Grambsch and Therneau 1994 Biometrika). Coefficients the violate the proportionality assumption were refit with time-dependendent covariates.

Results

PHS601 accurately stratifies Norwegian cohort based on PCa risk

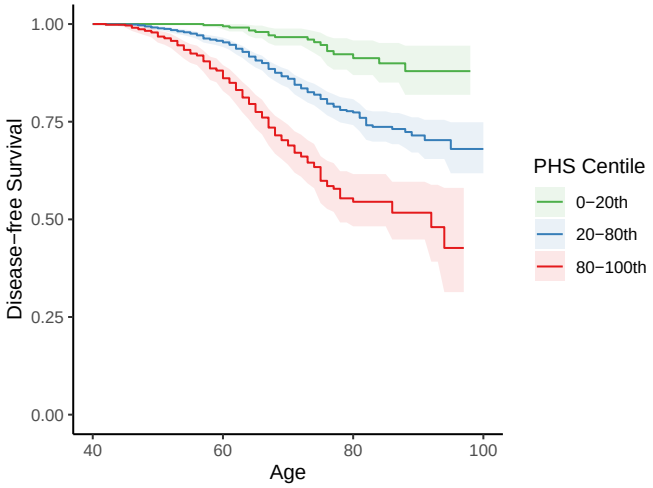


Figure 1: Kaplan-Meier survival curves of prostate cancer-free survival by centile ranges of polygenic hazard score. Time of "failure" is age at any diagnosis of prostate cancer. Controls were censored at age of last observation. Error ribbons are the 95% confidence interval.

In the NFCS cohort, there were 339 individuals with a PCa diagnosis and 2511 controls. The Cox proportional hazards model showed an 87% greater risk of PCa for a 1 standard deviation increase in PHS (HR = 1.87, 95% CI [1.67, 2.08], $p < 0.001$). Individuals with PHS in the top quintile had over 5 times greater risk of any PCa than individuals in the bottom quintile ($HR_{80/20}$: 5.72, 95% CI [4.30, 7.73]) and 139% greater risk than the median individual ($HR_{80/60}$: 2.39, 95% CI [2.10, 2.77]; Figure 1)

The proportionality test and plot of weighted residuals revealed a negative trend in HR with age suggesting that the proportionality assumption was invalid for PHS (Chi. sq. = 4.79, df = 1, p = 0.029). This can be seen by the decreasing OR at age 60, 70, and 80. The effect size seemed to decrease roughly linearly with age so we refit the model as an extended Cox model with a time-dependent PHS covariate with a linear time transform and a y-intercept at age 40. In this model, there was a 145% increase in risk for a 1 standard deviation increase in PHS at age 40 (HR = 2.45, 95% CI [1.86, 3.22], $p < 0.001$), which declined by

1.1% per year (HR = 0.99, 95% CI [0.98, 1.00], p = 0.035). However, the time varying effect size was weak and the weighted residual plot showed a wide spread demonstrating high variability in the time-varying coefficient. Therefore, this effect is likely not clinically significant and so in the previous paragraph and in Figure 1 we have presented the Cox model without the time-transformed PHS.

PHS601 displays specificity for metastatic PCa

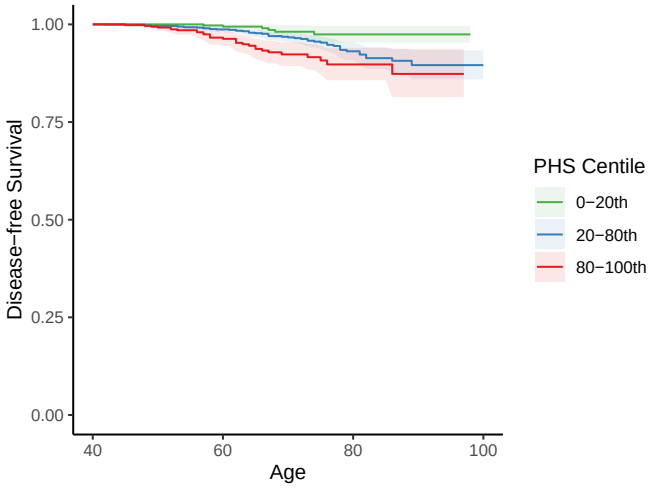


Figure 2: Kaplan-Meier survival curves of metastatic prostate cancer-free survival by centile ranges of polygenic hazard score. Time of "failure" is age at diagnosis of metastatic prostate cancer. Controls were censored at age of last observation. Error ribbons are the 95% confidence interval.

In addition to evaluating the performance of PHS601 in stratifying individuals based on risk of any PCa, we were also interested in the specificity of PHS601 for aggressive disease. In the NFCS cohort, 85 individuals had been diagnosed with metastatic disease leaving 2765 controls. The Cox proportional hazards model of metastatic disease showed a 72% greater risk of metastatic disease for a 1 standard deviation increase in PHS (HR = 1.72, 95% CI [1.39, 2.14], $p < 0.001$). Individuals with PHS in the top quintile had over 4 times greater risk of metastatic disease than individuals in the bottom quintile ($HR_{80/20}$: 4.57, 95% CI [2.71, 7.85]) and 114% greater risk than the median individual ($HR_{80/60}$: 2.14, 95% CI [1.66, 2.85]; Figure 2)

PHS and HOXB13 independently contribute to increased PCa risk

We compared the performance of RPVs and PHSes in predicting any PCa. Only the NFCS cohort received whole-genome sequencing so the HUSK cohort was excluded from these analyses. We first subsetted our dataset to include only subjects that received both whole genome sequencing and genotyping. We then re-analyzed the PHS data on this smaller subsample to compare it to the performance of the rare pathogenic variants. There were a

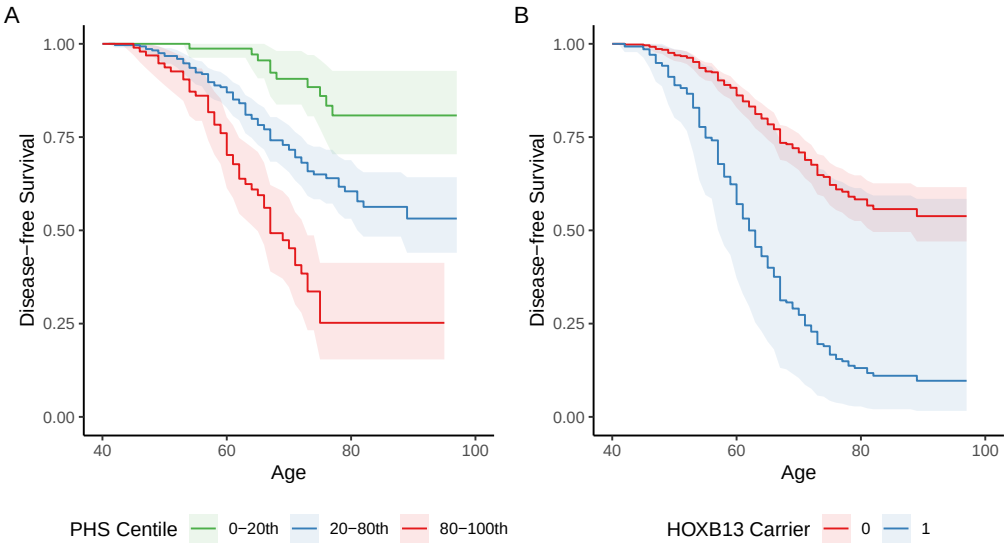


Figure 3: Kaplan-Meier survival curves of prostate cancer-free survival for the whole-genome sequenced sub-sample for **(A)** centile ranges of polygenic hazard score and **(B)** HOXB13 carriers. Time of "failure" is age at any diagnosis of prostate cancer. Controls were censored at age of last observation. Error ribbons are the 95% confidence interval.

total of 503 subjects with whole-genome sequencing of which 138 were cases and 365 were controls. Among these individuals, a total of 22 had a deleterious mutation in one of the seven genes that we investigated. No subjects were observed with mutations in MSH2 or MSH6 so these genes were removed from the analysis.

As a preliminary analysis, we tested whether the frequency of mutations was greater in cases than in controls for each gene using a one-sided Fisher's exact test. Only the HOXB13 gene showed increased frequency in the cases than in the controls (OR = 9.65, 95% lower confidence bound = 2.23, adj. p = 0.01). Therefore, we only proceeded with testing the risk of diagnosis with this gene. The HOXB13 gene mutation was at a frequency of 1.8% in the WGS subsample and having a high polygenic score does not appear to be associated with being more likely to carry a rare variant (OR = 1.65, 95% CI [0.85, 3.19], p = 0.14).

Using the subsetted WGS sample, the Cox proportional hazards model for PHS showed an 80% greater risk of PCa for a 1 standard deviation increase in PHS (HR = 1.80, 95% CI [1.53, 2.13], p < 0.001). Individuals with PHS in the top quintile had over 5 times greater risk of any PCa than individuals in the bottom quintile ($HR_{80/20}$: 5.09, 95% CI [3.35, 8.27]) and 128% greater risk than the median individual ($HR_{80/60}$: 2.28, 95% CI [1.86, 2.90]; Figure 5A). By comparison, the Cox proportional hazards model for HOXB13 carriers showed a 3.8 times greater risk of PCa for carriers of HOXB13 mutations than non-carriers (HR = 3.77, 95% CI [1.75, 8.11], p < 0.001; ; Figure 5B).

Combining these two predictors into a single Cox model, we find that both PHS and HOXB13 carrier status can independently contribute to increased PCa risk. In the combined main effect model, a 1 standard deviation increase in PHS led to a 79% increase PCa risk (HR = 1.79, 95% CI [1.51, 2.12], p < 0.001) while carrying a HOXB13 mutation increase risk by 2.9 times (HR = 2.91, 95% CI [1.35, 6.27], p = 0.006). The interaction term was not significant (HR = 0.85, 95% CI [0.26, 2.84], p = 0.79), therefore we have presented only the main effect model here.

We repeated this analysis by imputing the dataset using the WGS samples as an interal reference panel and identified individuals with HOXB13 SNPs in the full data set. The Cox proportional hazards model for HOXB13 SNP carriers showed a 4.3 times greater risk of PCa for carriers of the HOXB13 SNP than non-carriers (HR = 4.33, 95% CI [2.37, 7.90], p < 0.001; Figure 4).

PHS601 versus PHS290

Is this an improvement over the previous model?

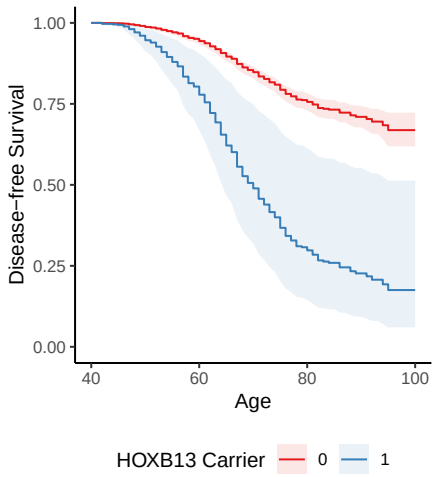


Figure 4: Kaplan-Meier survival curves of prostate cancer-free survival for HOXB13 SNP carriers from imputed genotype data. Time of "failure" is age at any diagnosis of prostate cancer. Controls were censored at age of last observation. Error ribbons are the 95% confidence interval.

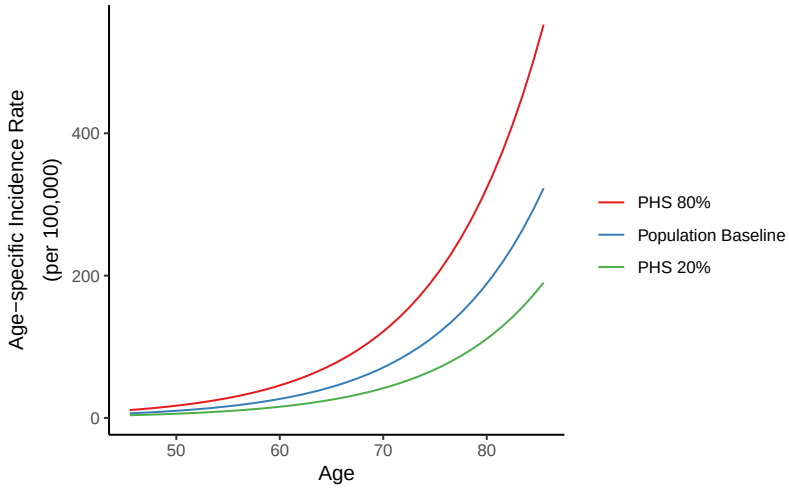


Figure 5: Cumulative incidence rate of metastatic prostate cancer per 100,000 individuals for the general population of Norway from the Norwegian Cancer Registry. Population baseline is for prostate cancer cases in Norway, 2014-2017 as reported by Huynh-Le et al. (2020). Incidence rates are stratified by the HR for the top 20% and bottom 20% of PHSes from the PHS601 model fit with data from the NFCS cohort.

Plots of incidence rate in the cohorts versus the general Norwegian population from the cancer registry?

Discussion

The strong stratification in the HUSK cohort demonstrates good risk stratification in the general population. In addition, the NFCS cohort was recruited based on self-reported family history of cancer and showed elevated cancer-risk relative to the general Norwegian population. Therefore, the strong risk stratification in both HUSK and the NFCS cohorts demonstrates that this model not only performs good risk stratification in the general population, but also in people with a family history or with concerns about cancer. Even in a group with a family history of cancer, there is a huge stratification between low and high risk. This suggests that genetics isn't just a proxy for family history.

Applications/Public Helpful

The benefit of this approach is that I can tell you before I have screened you how likely you are to develop disease and how likely you are to develop agressive/metastatic/clinically significant disease. You should know this before you decide whether to do measure PSA.

In addition, compared to screening for BRCA2 or even HOXB13, PHS has a bigger public health benefit in this case. Even in the case of low numbers, for example, in the NFCS case we had only 300 cases of prostatic cancer and 80 cases of metastatic disease, the polygenic score was more useful than sequencing in the same sample size. If BRAC2 is not useful for stratifying risk in a popualtion of 3000, how useful is it in the clinic when looking at a single patient? In addition, genotyping for polygenic scoring costs a fraction of sequencing (source? numbers on this?). A much larger percentage of the population are affected by a high risk score than are affected by currently known PRVs. High polygenic risk affects 50x more people than BRCA2 mutations, which means genetic risk scoring is more useful for more people. If someone is carrier of a pathogenic mutation in BRCA2 or HOXB13, that tells you they have a relatively high risk of developing cancer, however, it provides no information about the relative risk of patients without those rare variants. Whereas, a polygenic risk score can inform risk in all patients. Evaluating how this could work in Norway, there is more value per NOK for PHS screening than for sequencing.

It is possible that clinical interventions differ between individuals whose phenotype arises from rare deleterious mutations and individuals whose phenotype arises from the interaction of environment with genotype of the normal array of segregating genes of minor effect. Furthermore, RPVs and PHSes have different implications for genetic counseling for the risk of offspring being affected in the case of RPVs.

This suggests that both the normal array of segregating genes as well as rare deleterious mutations contribute independently to PCa risk. As we show, having a high PHS score does not appear to predict being a carrier of HOXB13 G84E. Therefore, both of these genetic factors can provide valuable information about

Future Directions

One limitation of this study is the limited sample size of subjects with diagnosis data, age of diagnosis, and metastasis code. If we could get a bigger data set, we could potentially identify additional SNPs specific to the Norwegian population.

While information on metastatic or aggressive disease is often lacking in study populations and sample sizes for metastatic disease are relatively small resulting in decreased power, advances in electronic health records systems and biobanks could improve the sample sizes and information necessary to identify individuals with metastatic disease. Therefore, future studies should develop models that are specific to the development of metastatic or aggressive disease

Acknowledgements

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References